

1. Define rational numbers.

The number of the form $\frac{p}{q}$ where p, q integers and $q \neq 0$ are called rational numbers. For example, $\frac{2}{9}, 7, \sqrt{\frac{25}{16}}$

$$Q = \left\{ x \mid x = \frac{p}{q}, \quad p, q \in \mathbb{Z} \wedge q \neq 0 \right\}$$

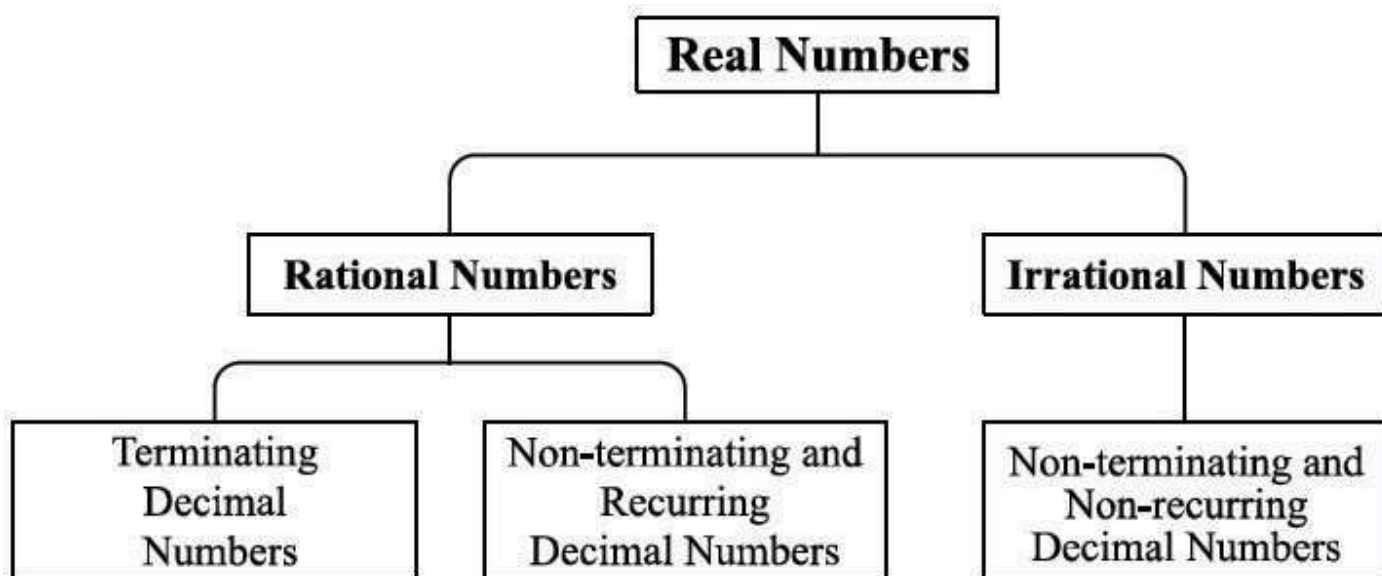
2. Define irrational numbers.

The number which cannot be expressed of the form $\frac{p}{q}$ where p, q integers and $q \neq 0$ are called irrational numbers. For example, $e, \pi, \sqrt{2}$

$$Q' = \left\{ x \mid x \neq \frac{p}{q}, \quad p, q \in \mathbb{Z} \wedge q \neq 0 \right\}$$

3. Define set of real numbers.

The union of the set of rational numbers and irrational numbers is known as the set of real numbers. It is denoted by R . i. e. $\mathbb{R} = Q \cup Q'$

**4. Explain decimal representation of rational number?**

The decimal representations of rational numbers are of two types.

- (i) Terminating Decimal Fractions
- (ii) Recurring and Non-Terminating Decimal Fractions

(i) Terminating Decimal Fractions

The decimal number with a finite number of digits after the decimal point is called a terminating decimal number. For example,

$$\frac{2}{5} = 0.4 \text{ and } \frac{3}{8} = 0.375$$

(ii) Recurring and Non-Terminating Decimal Fractions

The decimal numbers with an infinitely repeating pattern of digits after the decimal point are called non-terminating and recurring decimal numbers. For example,

$$\begin{aligned} \frac{1}{3} &= 0.333 \dots = 0.\overline{3} \text{ (3 repeats infinitely)} \\ \frac{1}{6} &= 0.166 \dots = 0.1\overline{6} \text{ (6 repeats infinitely)} \\ \frac{22}{7} &= 3.142857142857 \dots = 3.\overline{142857} \text{ (the pattern 142857 repeats infinitely)} \end{aligned}$$

5. What are irrational numbers in terms of their decimal representation?

Decimal numbers that do not repeat a pattern of digits after the decimal point continue indefinitely without terminating.

Non-terminating and non-recurring decimal numbers are known as irrational numbers.

Examples:

- $\pi = 3.1415926535897932 \dots$
- $e = 2.71828182845904 \dots$
- $\sqrt{2} = 1.41421356237309 \dots$

Note:

- (i) $e = 2.7182 \dots$ is called *Euler's Number*.
- (ii) *Rational number + Irrational number = Irrational number.*
- (iii) *Rational number ($\neq 0$) $\times$ Irrational number = Irrational number.*
- (iv) *The **product of two irrational numbers** can be either **rational** or **irrational**, depending on the numbers involved.*

6. Explain the concept of radicals and radicands.

If n is a positive integer greater than 1 and a is a real number, then any real number x such that $x^n = a$ is called (radical) the n th root of a , and in symbols is written as

$$\begin{aligned}x^n &= a \\[x^n]^{1/n} &= [a]^{1/n} \\x &= a^{1/n} \\x &= \sqrt[n]{a}\end{aligned}$$

In the radical $\sqrt[n]{a}$, the symbol $\sqrt{}$ is called the **radical sign**, n is called the **index** of the radical and the real number a under the radical sign is called the **radicand** or **base**.

7. Define surd.

An irrational radical with rational radicand is called a surd. For example,

- $\sqrt{5}$ is a surd because $\sqrt{5}$ does not give a whole number.
- $\sqrt{9}$ is **not** a surd because it simplifies to 3 (a whole number).
- $\sqrt{3}, \sqrt[3]{7}$ are surds.
- $\sqrt{\pi}$ and $\sqrt{e}$ are **not** surds.

Note: Every **surd** is an **irrational number**, but not every **irrational number** is a **surd** (e.g., $\sqrt{\pi}$ is irrational but not a surd) and the **product of two conjugate surds** is a **rational number**.

8. Define monomial surd.

A surd which contains a single term is called a monomial surd. For example, $\sqrt{2}, \sqrt{3}$

9. Define binomial surd.

A surd which contains sum of two monomial surds or sum of a monomial surd and a rational number is called a binomial surd. For example, $\sqrt{2} + \sqrt{7}, \sqrt{2} + 5$

10. Define conjugate surd.

Conjugate surd of $\sqrt{a} + \sqrt{b}$ is defined as $\sqrt{a} - \sqrt{b}$.

11. What are the additive properties of real numbers?

Name of the Property	$\forall a, b, c \in \mathbb{R}$	Examples
Closure	$a + b \in \mathbb{R}$	$2 + 3 = 5 \in \mathbb{R}$

Commutative	$a + b = b + a$	$2 + 5 = 5 + 2$ $7 = 7$
Associative	$a + (b + c) = (a + b) + c$	$2 + (3 + 5) = (2 + 3) + 5$ $2 + 8 = 5 + 5$ $10 = 10$
Identity	$a + 0 = a = 0 + a$	$5 + 0 = 5 = 0 + 5$
Inverse	$a + (-a) = (-a) + a = 0$	$6 + (-6) = (-6) + 6 = 0$

12. What are the multiplicative properties of real numbers?

Name of the Property	$\forall a, b, c \in \mathbb{R}$	Examples
Closure	$ab \in R$	$2 \times 5 = 10 \in R$
Commutative	$ab = ba$	$2 \times 5 = 5 \times 2$ $10 = 10$
Associative	$a(bc) = (ab)c$	$2 \times (3 \times 5) = (2 \times 3) \times 5$ $2 \times 15 = 6 \times 5$ $30 = 30$
Identity	$a \times 1 = a = 1 \times a$	$5 \times 1 = 5 = 1 \times 5$
Inverse	$a \times \frac{1}{a} = \frac{1}{a} \times a = 1$	$6 \times \frac{1}{6} = \frac{1}{6} \times 6 = 1$

Note:

- (i) 0 and 1 are the **additive** and **multiplicative** identities of real numbers, respectively.
- (ii) $0 \in R$ has no multiplicative inverse.

13. What are the distributive properties of real numbers?

Property	Mathematical Expression
Left Distributive Property of Multiplication over Addition	$a(b + c) = ab + ac$
Left Distributive Property of Multiplication over Subtraction	$a(b - c) = ab - ac$
Right Distributive Property of Multiplication over Addition	$(a + b)c = ac + bc$
Right Distributive Property of Multiplication over Subtraction	$(a - b)c = ac - bc$

14. What are the properties of equality of real numbers?

No.	Property Name	Mathematical Expression
i	Reflexive Property	$\forall a \in R,$ $a = a$
ii	Symmetric Property	$\forall a, b \in R,$ $a = b$ $\Rightarrow b = a$
iii	Transitive Property	$\forall a, b, c \in R,$ $a = b \wedge b = c$ $\Rightarrow a = c$
iv	Additive Property	$\forall a, b, c \in R,$ $a = b$ $\Rightarrow a + c = b + c$
v	Multiplicative Property	$\forall a, b, c \in R,$ $a = b$ $\Rightarrow ac = bc$

vi	Cancellation Property w.r.t Addition	$\forall a, b, c \in R,$ $a + c = b + c$ $\Rightarrow a = b$
vii	Cancellation Property w.r.t Multiplication	$\forall a, b, c \in R,$ $ac = bc$ $\Rightarrow a = b$

15. State and explain the Multiplicative Property of order with examples.

No.	Property Name	Mathematical Expression
i	Trichotomy Property	$\forall a, b \in R,$ Either $a = b$ or $a > b$ or $a < b$
ii	Transitive Property	$\forall a, b, c \in R,$ $\begin{aligned} \text{➤ } a > b \wedge b > c &\Rightarrow a > c \\ \text{➤ } a < b \wedge b < c &\Rightarrow a < c \end{aligned}$
iii	Additive Property	$\forall a, b, c \in R,$ $\begin{aligned} \text{➤ } a > b &\Rightarrow a + c > b + c \\ \text{➤ } a < b &\Rightarrow a + c < b + c \end{aligned}$
iv	Multiplicative Property	$\forall a, b, c \in R,$ $\begin{aligned} \text{➤ } a > b &\Rightarrow ac > bc \text{ if } c > 0 \\ \text{➤ } a < b &\Rightarrow ac < bc \text{ if } c > 0 \\ \text{➤ } a > b &\Rightarrow ac < bc \text{ if } c < 0 \\ \text{➤ } a < b &\Rightarrow ac > bc \text{ if } c < 0 \\ \text{➤ } a > b \wedge c > d &\Rightarrow ac > bd \\ \text{➤ } a < b \wedge c < d &\Rightarrow ac < bd \end{aligned}$
v	Division Property	$\forall a, b, c \in R,$ $\begin{aligned} \text{➤ } a > b &\Rightarrow \frac{a}{c} > \frac{b}{c} \text{ if } c > 0 \\ \text{➤ } a < b &\Rightarrow \frac{a}{c} < \frac{b}{c} \text{ if } c > 0 \\ \text{➤ } a > b &\Rightarrow \frac{a}{c} < \frac{b}{c} \text{ if } c < 0 \\ \text{➤ } a < b &\Rightarrow \frac{a}{c} > \frac{b}{c} \text{ if } c < 0 \end{aligned}$
vi	Reciprocal Property	$\forall a, b, c \in R$ and have same sign $\begin{aligned} \text{➤ } a > b &\Rightarrow \frac{1}{a} < \frac{1}{b} \\ \text{➤ } a < b &\Rightarrow \frac{1}{a} > \frac{1}{b} \end{aligned}$

16. Write the Laws of Radicals and Indices.

Laws of Radical	Laws of Indices
(i) $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$	(i) $a^m \cdot a^n = a^{m+n}$
(ii) $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$	(ii) $(a^m)^n = a^{mn}$
(iii) $\sqrt[n]{a^m} = (\sqrt[n]{a})^m$	(iii) $(ab)^n = a^n b^n$
(iv) $(\sqrt[n]{a})^n = \left(a^{\frac{1}{n}}\right)^n = a$	(iv) $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
	(v) $\frac{a^m}{a^n} = a^{m-n}$
	(vi) $a^0 = 1$

17. Is 0 a rational number? Explain.

Yes, 0 is a rational number because it can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

For example, $\frac{0}{1}$, $\frac{0}{2}$, $\frac{0}{3}$, etc., are valid rational numbers. Since 0 divided by any nonzero integer is always 0, it satisfies the definition of a rational number.

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